# Amazon Keys

1. Customer
Attributes: customer_id, Name, Email, Address, Phone, Date_of_birth

- PK : customer_id
- CK : customer_id
- SK : Any combination of attributes that includes a Candidate Key.
  Examples: {customer_id},{customer_id, Name},

2. Product
Attributes: product_id, Name, category_id, Description
- PK : product_id
- CK : product_id
- SK : Any combination containing the product_id.
  Examples : {product_id, Name}

3. Order
Attributes: order_id, customer_id, order_date, Status
- PK : order_id
- CK : order_id
- SK : Any combination containing order_id.
  Examples: {order_id}, {order_id, customer_id}, {order_id, Status, order_date}

4. Order_item
Attributes:order_id, product_id, seller_id, Quantity, unit_price, Shipment-id
- PK : {order_id, product_id, seller_id}
- CK : {order_id, product_id, seller_id}
- SK : Any combination containing both order_id and product_id and seller_id.

5. Seller
Attributes: seller_id, store_name, contact_num
- PK : seller_id
- CK : seller_id, store_name (if store names must be unique), and contact_num (if unique to each seller).
- SK : Any combination containing a Candidate Key.
  Examples: {seller_id}, {seller_id, Rating}, {store_name, contact_num}

6. Review
Attributes: review_id, product_id, customer_id, Rating, Comment, review_date
- PK : review_id
- CK : review_id
- SK : Any combination containing review_id.

Examples: {review_id}, {review_id, Rating}, {review_id, product_id, customer_id}

7. Payment
Attributes: payment_id, order_id, payment_method, Amount, payment_date, status
- PK : payment_id
- CK : payment_id
- Super Keys: Any combination containing payment_id.

Examples: {payment_id}, {payment_id, Amount, payment_method}

8. Shipment
Attributes: shipment_id, warehouse- id,delivery_date, shipping_address, Status
- PK : shipment_id
- CK : shipment_id
- SK :Any combination containing shipment_id.

Examples:{shipment_id, Status}, {shipment_id, delivery_date}
9. Worker
Attributes: worker_id, Name, phone_num, hire_date,Date_of_birth,Salary, Role
- PK : worker_id
- CK : worker_id , phone_num
- SK : Any combination containing a Candidate Key.

Examples: {worker_id}, {phone_num}, {worker_id, Role, Salary}

10. Warehouse
Attributes: warehouse_id, manager_id, Location, Capacity
- PK : warehouse_id
- CK : warehouse_id
- SK : Examples: {warehouse_id}, {warehouse_id, manager_id, Capacity}

11. Product-Seller
Attributes:Product-id, Seller-id , price
- PK :{Product-id, Seller-id}
- CK :{Product-id, Seller-id}
- SK : Examples:{Product-id, Seller-id} , {Product-id, Seller-id, Price}

12. Inventory
Attributes: warehouse_id, Product-id, Seller-id, Quantity
- PK : {warehouse_id, Product-id, Seller-id}
- CK : {warehouse_id, Product-id, Seller-id}
- SK : Examples:{warehouse_id, Product-id, Seller-id, Quantity}

13.Category
Attributes:category_id, category_name
- PK :category_id
- CK : category_id, category_name
- SK : {category_id, category_name},…

## Foreign Keys

1. Product_Seller.product_id → Product.product_id

2. Product_Seller.seller_id → Seller.seller_id

3. Inventory.warehouse_id → Warehouse.warehouse_id

4. Inventory.{product_id, seller_id} → Product_Seller.{product_id, seller_id} (Composite foreign key linking warehouse stock directly to a specific merchant's listing).

5. Warehouse.manager_id → Worker.worker_id

6. Order.customer_id → Customer.customer_id

7. Order_item.order_id → Order.order_id

8. Order_item.{product_id, seller_id} → Product_Seller.{product_id, seller_id}

9. Order_item.shipment_id → Shipment.shipment_id (Allows multi-item orders to be split across different fulfillment boxes).

10. Shipment.warehouse_id → Warehouse.warehouse_id

11. Review.product_id → Product.product_id

12. Review.customer_id → Customer.customer_id

13. Payment.order_id → Order.order_id

14. Product.category_id → Category.category_id

## Relationships

- Product (1) ——— (M) Product_Seller
- Seller (1) ——— (N) Product_Seller
- Warehouse (1) ——— (P) Inventory

A physical `Warehouse` logistical building can store and track quantities for many unique product-seller inventory items.

- Product_Seller (1) ——— (Q) Inventory

A specific seller's product listing (`Product_Seller`) can have its physical stock distributed across multiple different regional warehouses to optimize delivery routing.

- Customer (1) ——— (N) Order
- Order (1) ——— (M) Order_item
- Product_Seller (1) ——— (N) Order_Item
- Shipment (1) ——— (N) Order_item
- Warehouse (1) ——— (N) Shipment
- Order (1) ——— (N) Payment
- Product (1) ——— (N) Review
- Customer (1) ——— (N) Review
- Category (1) ——— (N) Product